

NON-LINEAR MULTI-FRACTAL AND WAVELET ANALYSIS OF TEMPORAL VARIATIONS OF THE MARKET STOCK INDEXES IN CONDITIONS OF THE GLOBAL ECONOMY CRISIS

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The purpose of this paper is to carry out a non-linear, multi-fractal and wavelet analysis of the temporal variations of the market stock indexes (Russia's RTC, MFCB, Germany's Xetra Dax index, PE FTSE Eurofirst 300, Nikkei 225 S&P500 etc.) during the quite long temporal interval (including seven and more years per day, week, month; including the period of global economy crisis 2008 -...) and to find the corresponding multi-fractal features and signatures of the low-dimensional chaos. Non-linear prediction model is introduced and applied to the predicting the behavior of the cited market stock indexes.

Since last decades, many scientists use the new powerful tool based on the wavelet decomposition for analyzing various signals series, including financial ones [1]. At present, the family of analyzing function dubbed wavelets is being increasingly used in problems of pattern recognition, financial analysis; in processing and synthesizing various signals; in analysis of images of any kind [1-5]. Wavelets are fundamental building block functions, analogous to the sine and cosine functions. Fourier transform extracts details from the signal frequency, but all information about the location of a particular frequency within the signal is lost. At the expense of their locality the wavelets have advantages over Fourier transform when non-stationary signals are analyzed.

Wavelet analysis. Here, we use non-decimated wavelet transform that has temporal resolution at coarser scales and allows to isolate time series of the major components of financial sets a direct way [1-3]. The dilation and translation of the mother wavelet $\psi(t)$ generates the wavelet as follows: $\Psi_{j,k} = 2^{j/2} \psi(2^j t - k)$. The dilation parameter j controls how large the wavelet is, and the translation parameter k controls how the wavelet is shifted along the t -axis. For a suitably chosen mother wavelet $\psi(t)$, the set $\Psi_{j,k}$ provides an orthogonal basis. The corresponding wavelet expansion of a function is closely related to the discrete wavelet transform of a signal observed at discrete points in time. In practice, the length of the signal, say n , is finite and, for our study, the data are available daily, monthly, i.e. the interested function is vector $f = (f(t_1); \dots; f(t_n))$ with $t_i = i/n$ and $i = 1, \dots, n$. With these notations, the DWT of a vector f is simply a matrix product $d = W \times f$, where d is an $n \times 1$ vector of discrete wavelet coefficients indexed by 2 integers, d_{jk} , and W is an orthogonal $n \times n$ matrix associated with the wavelet basis. For computational reasons, it is simpler to perform the wavelet transform on time series of dyadic (power of 2) length. Using a link between wavelets and fractals, we have made calculating the multi-fractal spectrum.

Nonlinear prediction model. Before building a model of prediction, it's important to define how predictable is a chaotic system? The predictability can be estimated by the Kolmogorov entropy, which is proportional to a sum of positive Lyapunov exponents (LE). The spectrum of LE is one of

dynamical invariants for non-linear system with chaotic behaviour. The limited predictability of the chaos is quantified by the local and global LE, which can be determined from measurements. The LE are related to the eigenvalues of the linearized dynamics across the attractor. Negative values show stable behaviour while positive values show local unstable behaviour. For chaotic systems, being both stable and unstable, the LE indicate the complexity of the dynamics [4-7]. The largest positive value determines some "average" prediction limit. Since the LE are defined as asymptotic average rates, they are independent of the initial conditions, and hence the choice of trajectory, and they do comprise an invariant measure of the attractor. An estimate of this measure is a sum of the positive LE. The estimate of the attractor dimension is provided by the conjecture d_L and the LE are taken in descending order. The dimension d_L gives values close to the dimension estimates discussed earlier and is preferable when estimating high dimensions. To compute the LE, we use the method with linear fitted map, although the maps with higher order polynomials can be also used.

Non-linear model. Non-linear model of chaotic processes is based on the concept of compact geometric attractor on which observations evolve. Since an orbit is continually folded back on itself by dissipative forces and the non-linear part of dynamics, some orbit points $\mathbf{y}^r(k)$, $r=1, 2, \dots, N_B$ can be found in the neighbourhood of any orbit point $\mathbf{y}(k)$, at that the points $\mathbf{y}^r(k)$ arrive in the neighbourhood of $\mathbf{y}(k)$ at quite different times than k . One can then choose some interpolation functions, which account for whole neighbourhoods of phase space and how they evolve from near $\mathbf{y}(k)$ to whole set of points near $\mathbf{y}(k+1)$ [1]. The implementation of this concept is to build parameterized non-linear functions $\mathbf{F}(\mathbf{x}, \mathbf{a})$ which take $\mathbf{y}(k)$ into $\mathbf{y}(k+1) = \mathbf{F}(\mathbf{y}(k), \mathbf{a})$ and use various criteria to determine parameters $\mathbf{a}$. Since one has the notion of local neighbourhoods, one can build up one's model of the process neighbourhood by neighbourhood and, by piecing together these local models, produce a global non-linear model that capture much of the structure in an attractor itself.

The results of analysis are presented for temporal variations of the market stock indexes: Germany's Xetra Dax index, PE FTSE Eurofirst 300 one, Nikkei 225 one, S&P500, Russia's RTC, MBCB ones etc. Especial interest attracts an analysis of the data for cited indexes during the global economy crisis and future prediction data.

Keywords

temporal variations, market stock indexes, multi-fractals, wavelet analysis

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